Question 1: Archetype-Level Descriptive Statistics

Results and Interpretations

1 Research Question

How do age, height and weight distributions vary across broad sport categories and Athletics sub-disciplines, by sex where sample sizes allow? As a supplementary analysis, we also compare BMI and seven functional archetypes.

The analysis examines differences in central tendency, variability, distributional characteristics, and sex-wise anthropometric profiles across seven functional archetypes.

2 Sample Composition and Archetype Distribution

The dataset contains 737 athlete–year–archetype observations representing 522 unique athlete IDs, classified into seven functional archetypes. The 522 unique-athlete figure is a distinct count over all groups, not the sum of archetype counts.

Table 1: Distribution of Observations Across Archetypes

Archetype	**Observations**	**Unique Athletes**
Combat	110	84
Endurance	56	44
Precision	97	61
Racket Sports	61	36
Speed & Power	139	104
Team Sports	245	173
Technical / Mixed	29	24
Total	**737**	**522**

Note: Unique-athlete counts within archetypes are not additive because some athletes may belong to multiple archetypes.

Interpretation

Team Sports accounts for the largest share of observations (33.2%), followed by Speed & Power (18.9%) and Combat (14.9%).

Technical / Mixed has the smallest representation, contributing only 3.9% of observations.

This unequal distribution implies that descriptive estimates for larger archetypes are based on more observations, whereas estimates for smaller groups may be more sensitive to individual athletes.

Additionally, some athletes contribute in multiple Olympic years or archetypes. Therefore, the estimates describe retained athlete–year–archetype observations rather than a sample of independent athletes.

3 Relationship to Checkpoint 1 and Unit of Analysis

Checkpoint 1 specified comparisons of age, height and weight across broad sport categories, with an explicit Athletics sub-discipline breakdown (throws, sprints/hurdles, jumps, and distance running) and sex-specific comparisons where sample sizes allow. The present seven-archetype framework is an *additional analytical grouping*, not a replacement for the Athletics comparison in Section 5.

The cleaned input contains **835 event-level rows** for **522 athlete IDs**. The classification notebook assigns each row to a functional archetype using sport, and selected event names for Athletics, Swimming and Cycling. It then sorts and retains one row per `(ID, Year, Archetype)`. This consolidates **98 rows**, leaving **737 athlete–year–archetype observations**, **732 distinct athlete–year combinations**, and all **522 athlete IDs**. Thus, a participant may still contribute in multiple Olympic years or in two archetypes during the same year. The descriptive estimates are weighted by these retained observations and should not be interpreted as estimates from 737 independent athletes. No missing-data or statistical significance claims are implied by this consolidation.

For the event-specific Athletics comparison, the *pre-consolidation* 835-row cleaned file is used, because a broad archetype can contain more than one Athletics sub-discipline. Within Athletics, the analysis retains one row per `(ID, Year, Discipline)` to avoid counting multiple events in the same sub-discipline twice in the same edition. Athletes competing in different disciplines or different years can still contribute more than once. Age values include those previously imputed during cleaning; this should be kept in mind when interpreting age distributions.

4 Comparison of Mean Anthropometric Characteristics

Table 2: Mean Anthropometric Characteristics by Archetype

Archetype	**Age (years)**	**Height (cm)**	**Weight (kg)**	**BMI**
Combat	24.10	169.64	68.85	23.69
Endurance	26.21	172.54	61.04	20.36
Precision	29.21	171.80	71.57	24.07
Racket Sports	26.20	176.66	70.98	22.66
Speed & Power	24.90	171.25	68.06	22.85
Team Sports	24.74	171.80	66.20	22.41
Technical / Mixed	28.90	172.59	65.76	22.03

Interpretation

Precision athletes have the highest average age, weight, and BMI, while Racket Sports athletes are the tallest on average.

Endurance athletes exhibit the lowest mean weight and BMI, despite having an average height comparable to most other archetypes.

These observations suggest that physical profiles differ across functional categories.

However, these are descriptive comparisons and cannot establish that the demands of a particular sporting category cause the observed physical characteristics.

5 Athletics Sub-Discipline Comparison

To address the Athletics component explicitly promised in Checkpoint 1, Athletics entries are grouped into throws (discus, hammer, javelin, shot put), sprints/hurdles (100–400 m events and relays), jumps (high, long, triple), and distance running/walks (800 m and longer, marathon, and race walks). Decathlon and heptathlon are *combined events*, so they are excluded from the four-category comparison rather than assigned arbitrarily to one. Counts below use the one-athlete–year–discipline rule described earlier.

Table 3: Athletics sub-disciplines: descriptive means and sample composition

Discipline	n	Female	Male	Age (yr)	Height (cm)	Weight (kg)
Throws	27	11	16	27.96	183.00	99.33
Sprints / hurdles	61	37	24	24.31	168.62	59.07
Jumps	19	5	14	25.68	178.00	67.63
Distance / walks	45	13	32	26.87	169.78	57.96

The table compares group-level central tendencies. If the supplementary Athletics boxplots are provided, they additionally show within-group spread and atypical observations. The female and male counts indicate whether a within-sex interpretation is adequately supported; comparisons from small subgroups should remain descriptive. These results reflect observed Indian Olympic participation and should not be used to claim that sporting events cause body-size differences. The definitions of these four categories are tailored to the events actually represented in the cleaned data; in particular, the distance group includes race walks.

6 Age Distribution Across Functional Archetypes

Interpretation

Precision athletes have the highest mean age (29.21 years), closely followed by Technical / Mixed athletes (28.90 years).

Combat athletes have the lowest mean age (24.10 years), followed by Team Sports (24.74 years) and Speed & Power (24.90 years).

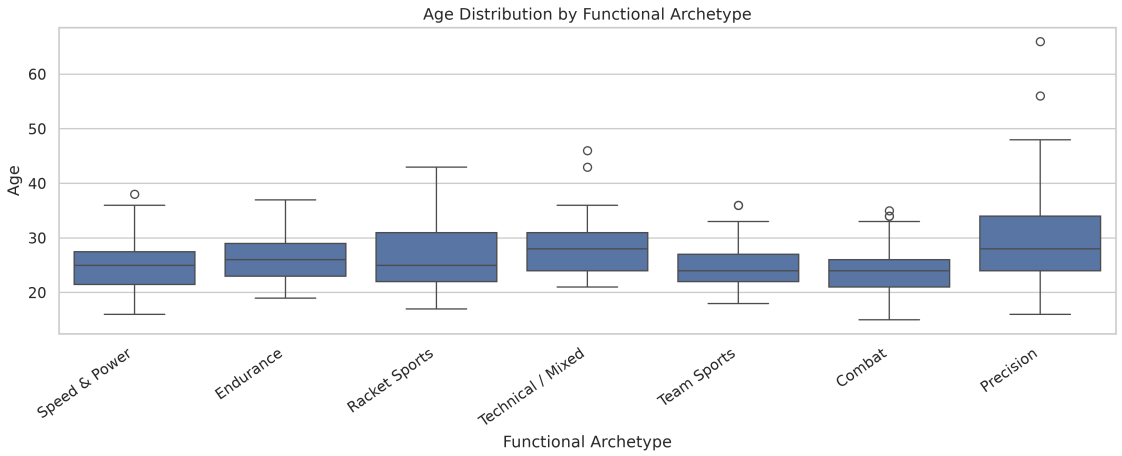


Figure 1: Age distributions by functional archetype.

The difference between the highest and lowest archetype means is approximately **5.11 years**, indicating noticeable variation in age profiles across the groups.

Precision also exhibits the greatest age variability, with a standard deviation of 8.57 years and an observed age range of 16–66 years.

This indicates a relatively heterogeneous age distribution, including considerably older participants.

By contrast, Team Sports has the smallest age standard deviation (3.60 years), suggesting that its observations are concentrated within a narrower age range.

The median age for Precision is 28 years, below its mean of 29.21 years.

Together with its high maximum age, this suggests a right-skewed distribution influenced by older athletes.

Key Finding: Precision and Technical / Mixed athletes tend to be older, while Combat and Team Sports exhibit younger and relatively concentrated age profiles.

7 Height Distribution Across Functional Archetypes

Interpretation

Racket Sports athletes have the highest average height (176.66 cm), approximately 7.02 cm greater than Combat athletes, who have the lowest mean height (169.64 cm).

Endurance, Precision, Speed & Power, Team Sports, and Technical / Mixed have mean heights between approximately 171 and 173 cm.

This indicates relatively small differences in average stature among these five groups.

Speed & Power has the largest height standard deviation (11.54 cm), suggesting substantial variation in body stature within this archetype.

This is consistent with the possibility that its constituent events include athletes with different physical requirements.

Technical / Mixed has the smallest height standard deviation (7.80 cm), although its relatively

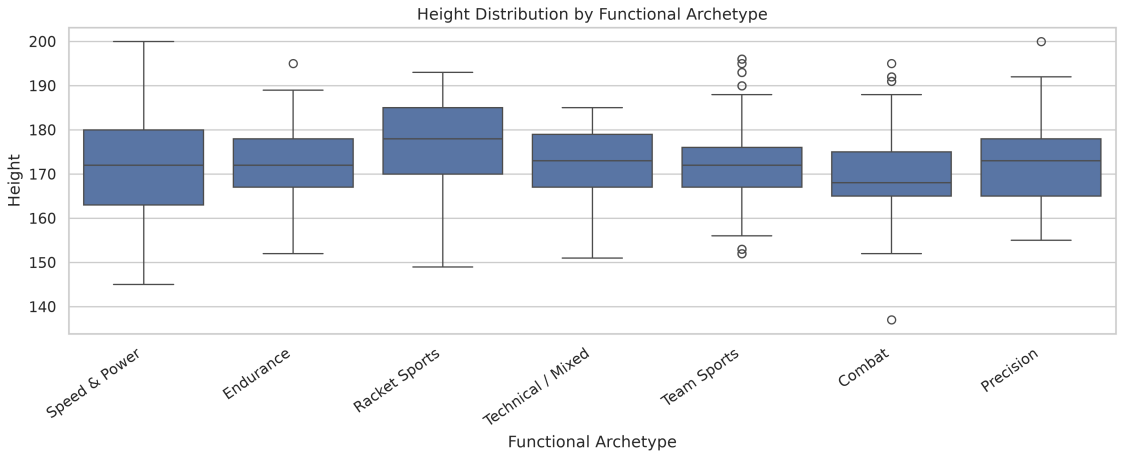


Figure 2: Height distributions by functional archetype.

small sample size should be considered when comparing variability.

The median height of Racket Sports athletes (178 cm) also exceeds that of the other groups.

This indicates that its relatively high mean is not merely the result of a few exceptionally tall observations.

Key Finding: Racket Sports shows a comparatively taller physical profile, whereas height differences among most other archetypes are relatively modest.

8 Weight Distribution Across Functional Archetypes

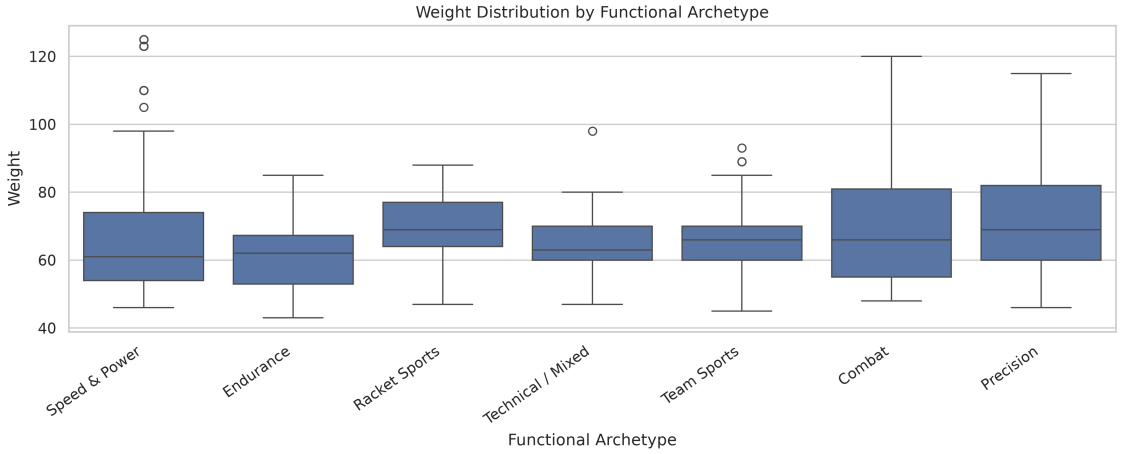


Figure 3: Weight distributions by functional archetype.

Interpretation

Precision athletes have the highest mean weight (71.57 kg), closely followed by Racket Sports (70.98 kg).

Endurance athletes have the lowest mean weight (61.04 kg).

The difference between the highest and lowest mean weights is approximately **10.53 kg**, demonstrating considerable variation in body mass across functional archetypes.

Speed & Power exhibits the greatest weight variability, with a standard deviation of 19.15 kg and weights ranging from 46 to 125 kg.

Its mean weight (68.06 kg) is substantially greater than its median (61 kg), indicating a right-skewed distribution in which heavier observations increase the mean.

Combat athletes also show substantial weight variability (SD = 16.47 kg), which may reflect differences in body size and weight-category participation.

By contrast, Team Sports has a comparatively low weight standard deviation (7.89 kg), suggesting a more concentrated weight distribution.

Endurance athletes maintain the lowest average weight despite having an average height of 172.54 cm, similar to several other archetypes.

Key Finding: Endurance athletes display a relatively lighter physical profile, whereas Speed & Power and Combat exhibit greater diversity in body mass.

9 BMI Distribution Across Functional Archetypes

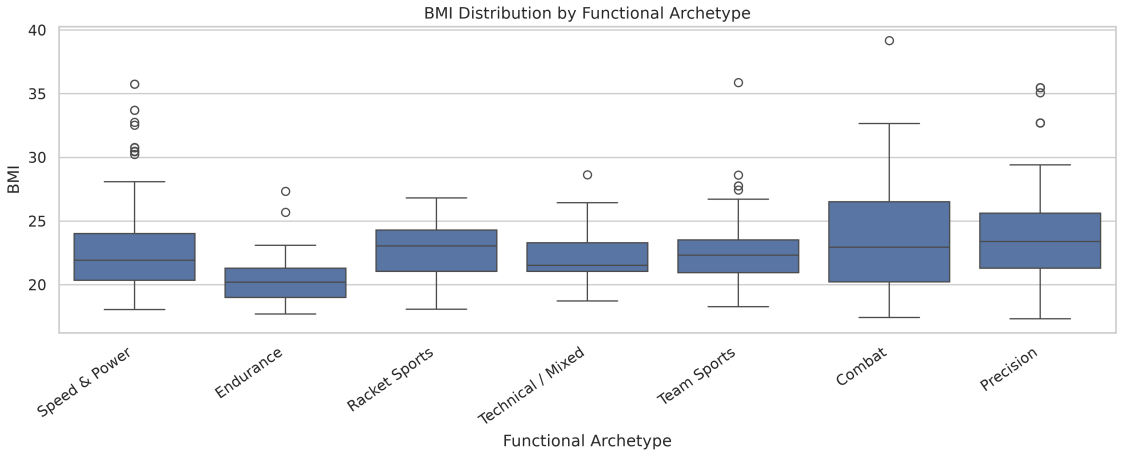


Figure 4: BMI distributions by functional archetype.

Interpretation

Precision athletes have the highest mean BMI (24.07), followed by Combat (23.69).

Endurance athletes have the lowest mean BMI (20.36).

The difference between the highest and lowest mean BMI values is approximately **3.71 kg/m^2**.

This indicates descriptive variation in body weight relative to height.

Endurance has the lowest BMI standard deviation (1.82), suggesting that its observations are clustered relatively closely around a lower BMI.

Racket Sports and Team Sports also exhibit comparatively concentrated BMI distributions.

Precision has the largest BMI standard deviation (4.32), followed by Combat (4.02) and Speed & Power (3.91).

This indicates greater variation in height-adjusted body mass within these groups.

The maximum recorded BMI in Combat is approximately 39.18, considerably above its group median of 22.94.

Similarly, Speed & Power and Precision contain relatively high BMI observations.

These values contribute to the observed upper tails and greater dispersion.

Importantly, BMI cannot distinguish muscle mass from body fat.

Therefore, a higher BMI among athletes should not be interpreted directly as greater adiposity or poorer physical fitness.

Key Finding: Endurance athletes show the lowest and most consistent BMI profile, while Precision, Combat, and Speed & Power exhibit greater heterogeneity in height-adjusted body mass.

10 Sex-Wise Comparisons

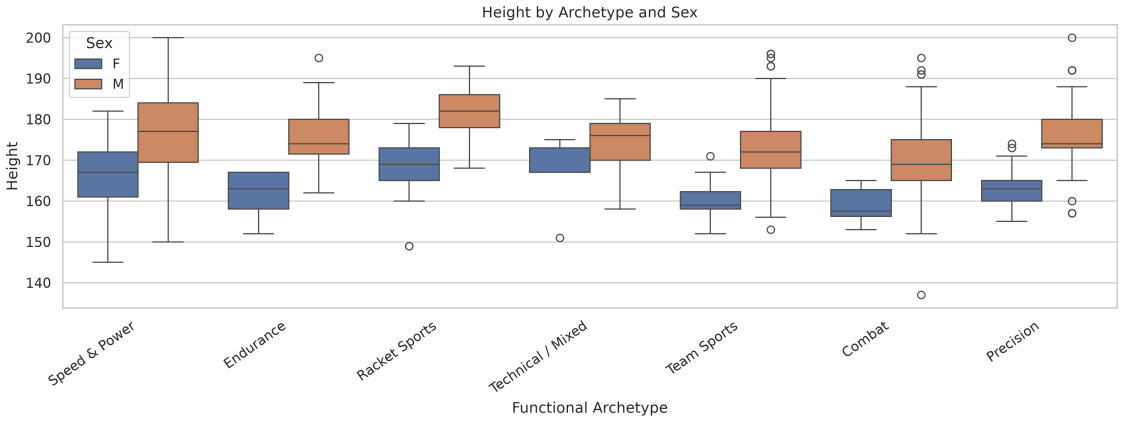


Figure 5: Height distributions by archetype and sex.

Sex-wise analysis is important because the relative proportions of male and female observations vary substantially between archetypes.

Table 4: Sex-Wise Comparison of Mean Height and BMI

Archetype	**Female Height**	**Male Height**	**Female BMI**	**Male BMI**
Combat	158.88	170.48	22.71	23.77
Endurance	162.00	175.72	18.63	20.89
Precision	162.86	175.43	21.90	24.95
Racket Sports	167.95	181.56	22.06	23.00
Speed & Power	165.78	175.79	21.53	23.94
Team Sports	159.94	172.62	21.22	22.49
Technical / Mixed	167.50	174.52	21.02	22.42

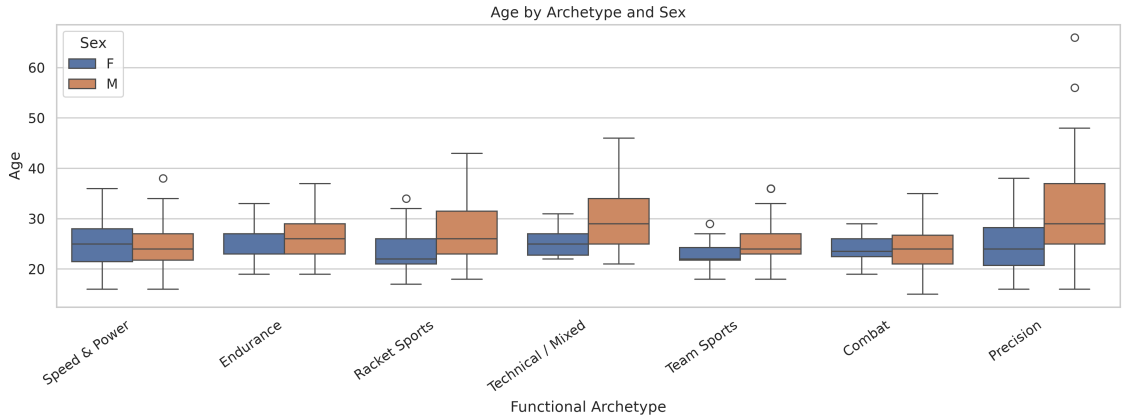


Figure 6: Age distribution by functional archetype and sex.

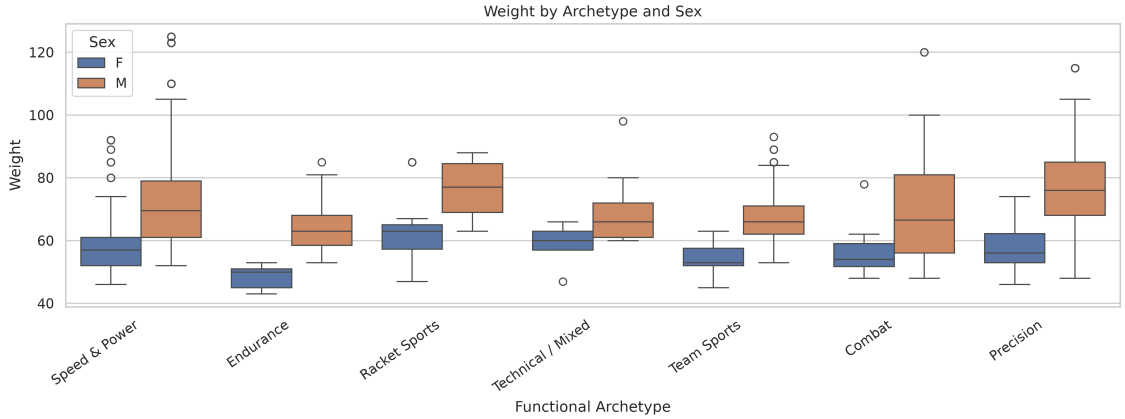


Figure 7: Weight distribution by functional archetype and sex.

10.1 Age Comparison

Within Precision, male observations have a substantially higher mean age (30.96 years) than female observations (24.89 years).

A similar pattern appears in Technical / Mixed, where males average 30.19 years compared with 25.50 years for females.

Conversely, Speed & Power has a slightly higher mean age among females (25.21 years) than males (24.64 years).

Endurance has almost identical mean ages for males and females.

These results indicate that differences in age composition between sexes are not uniform across archetypes.

10.2 Height Comparison

Male observations have a higher mean height than female observations across all seven archetypes.

The largest mean difference occurs in Endurance (13.72 cm), closely followed by Racket Sports (13.61 cm).

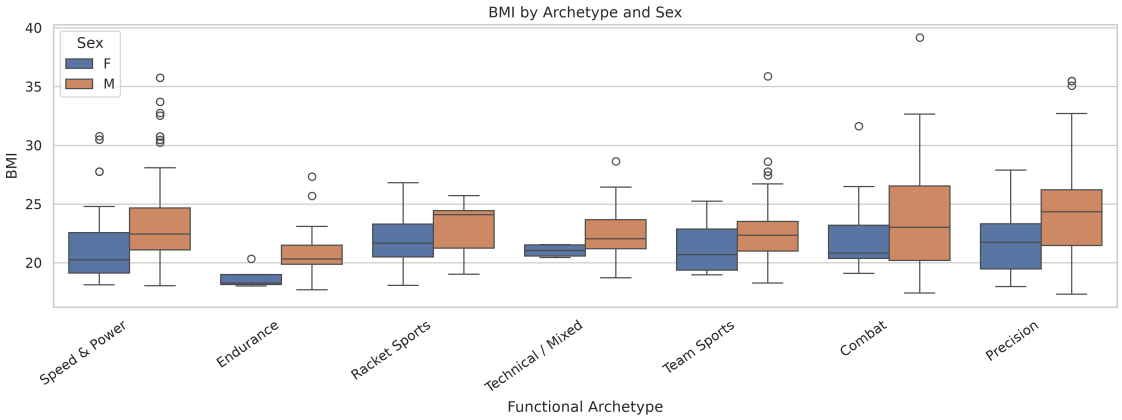


Figure 8: BMI distribution by functional archetype and sex.

Racket Sports has the highest mean height for both males (181.56 cm) and females (167.95 cm).

This indicates that its relatively tall profile persists when the sexes are analyzed separately.

10.3 Weight Comparison

Male observations have a higher mean weight across every archetype.

The largest sex-wise mean weight difference occurs in Precision, where males average 77.03 kg compared with 58.11 kg among females.

For both sexes, Endurance has the lowest mean weight: 64.70 kg for males and 48.92 kg for females.

This shows that the relatively low weight of Endurance athletes is not solely an artifact of the overall male-to-female composition.

10.4 BMI Comparison

Male observations have a higher mean BMI in every archetype.

The largest difference appears in Precision, where mean BMI is 24.95 for males and 21.90 for females.

Among female observations, Endurance has the lowest mean BMI (18.63), while Combat has the highest (22.71).

Among males, Endurance is again the lowest (20.89), while Precision is the highest (24.95).

Thus, the relatively low BMI of Endurance athletes persists for both sexes.

This strengthens the descriptive observation that this archetype has a lighter height-adjusted body-mass profile.

Important Limitation

Sex representation is highly unequal in several groups.

For example, Team Sports has 229 male observations but only 16 female observations, and Combat has 102 male observations compared with eight female observations.

These imbalances limit the reliability of direct comparisons, particularly within the smaller female subgroups.

11 Distributional Patterns from Histograms and Boxplots

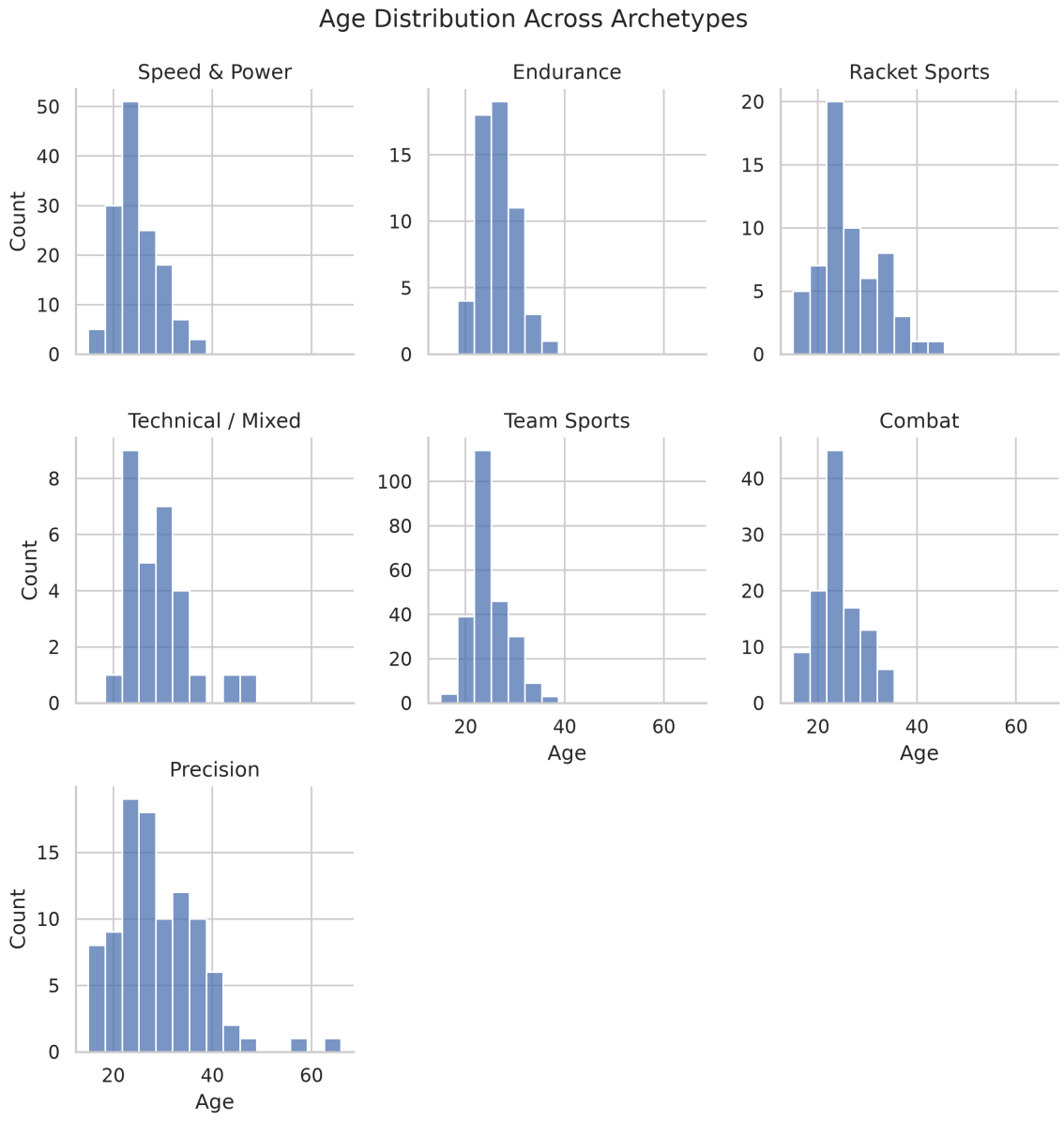


Figure 9: Age histograms for the seven archetypes.

The histogram and boxplot comparisons provide additional information beyond differences in group means.

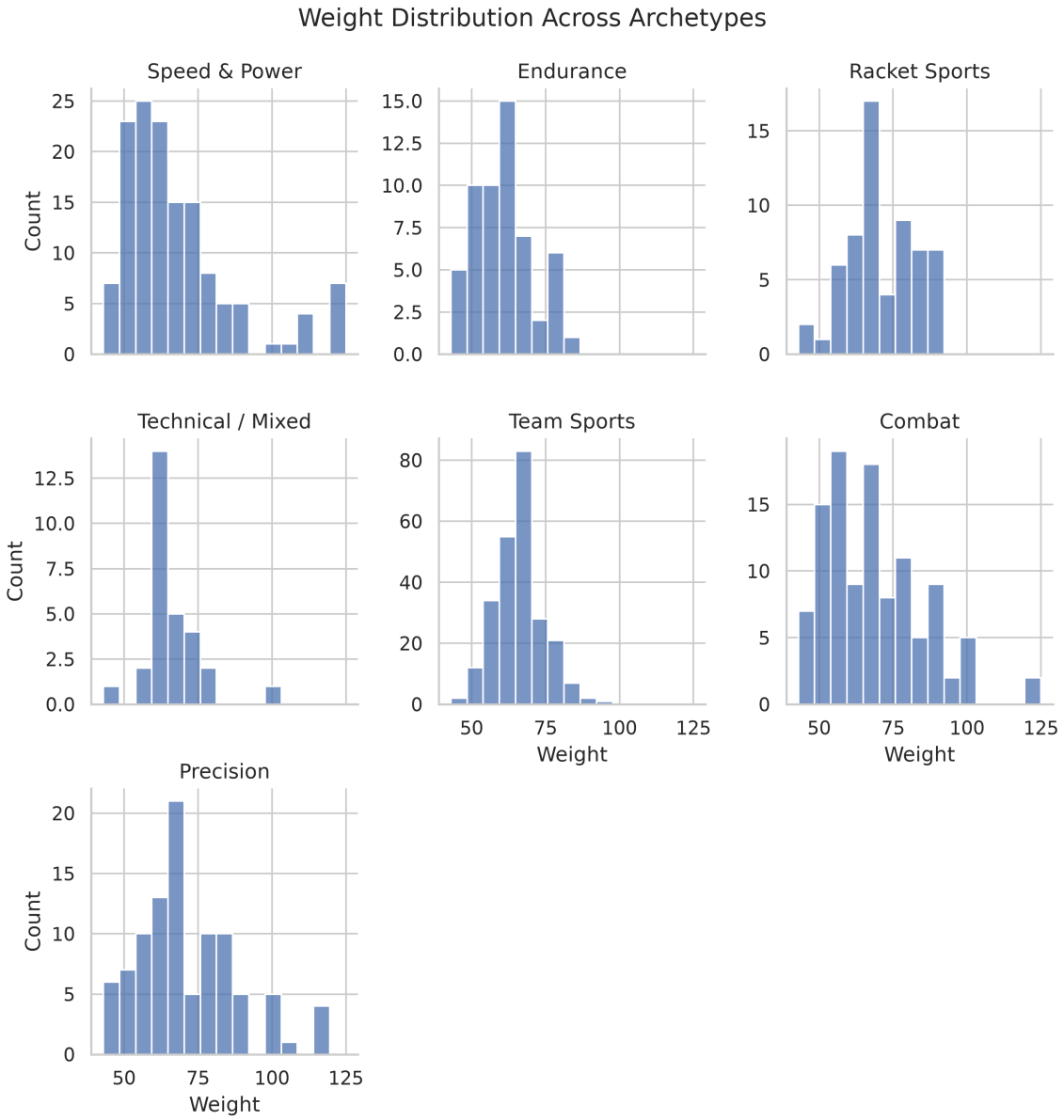


Figure 10: Weight histograms for the seven archetypes.

11.1 Age

Precision has the widest age distribution, including observations substantially older than those in most other groups.

Team Sports and Combat show comparatively concentrated age distributions.

The presence of older observations in Precision contributes to its higher mean and standard deviation.

11.2 Height

Most archetypes show considerable overlap in their height distributions, even though their means differ.

Racket Sports has a relatively high median height, while Speed & Power exhibits the widest interquartile range (17 cm).

This indicates substantial variation within the middle 50% of its height observations.

11.3 Weight

Speed & Power and Combat exhibit substantial variation in weight.

In Speed & Power, the difference between the mean and median is particularly pronounced, suggesting that heavier athletes contribute to an extended upper tail.

Team Sports has a narrower interquartile range (10 kg), indicating greater concentration in the central portion of its weight distribution.

11.4 BMI

Endurance has a relatively small BMI interquartile range (2.30), while Technical / Mixed has an even smaller IQR (2.24).

Combat has the largest BMI interquartile range (6.31), indicating substantial differences in height-adjusted weight even among the central 50% of observations.

The presence of extreme BMI values in certain archetypes reinforces the need to examine distributions rather than relying exclusively on mean values.

Overall, the archetypes differ not only in their average physical characteristics but also in the extent of within-group variation.

12 Height–Weight Relationship

The height–weight scatterplot examines whether taller athletes generally tend to have higher body weight.

The pooled observations show a moderately strong positive Pearson correlation:

$$\boxed{r = 0.698}$$

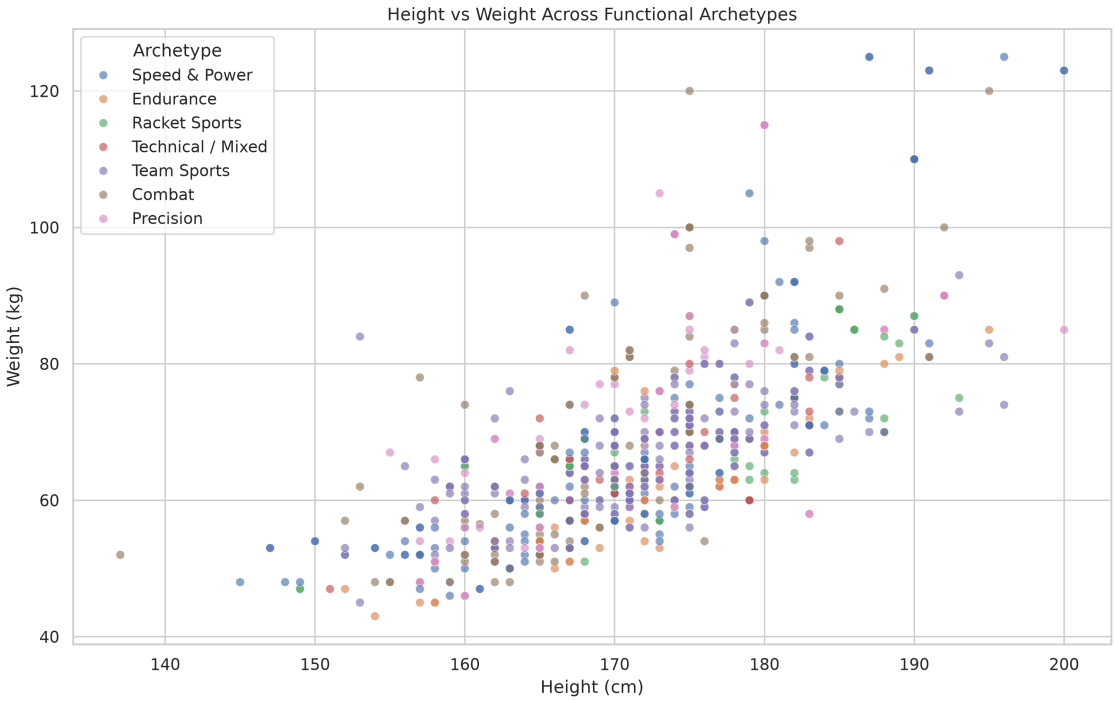


Figure 11: Height–weight relationship by archetype.

This indicates that taller athletes generally tend to be heavier, although substantial variation remains.

The strength of this association varies by archetype.

Endurance exhibits a particularly strong positive correlation:

$$r_{\text{Endurance}} = 0.882$$

Precision shows a weaker, though still positive, correlation:

$$r_{\text{Precision}} = 0.635$$

These differences suggest that the relationship between height and weight is not identical across functional groups.

However, these correlations do not establish causal relationships.

Pooled associations may also reflect differences in sex composition and sporting events.

13 Age–BMI Relationship

The overall Pearson correlation between age and BMI is approximately:

$$\boxed{r = 0.240}$$

This indicates a weak positive linear association.

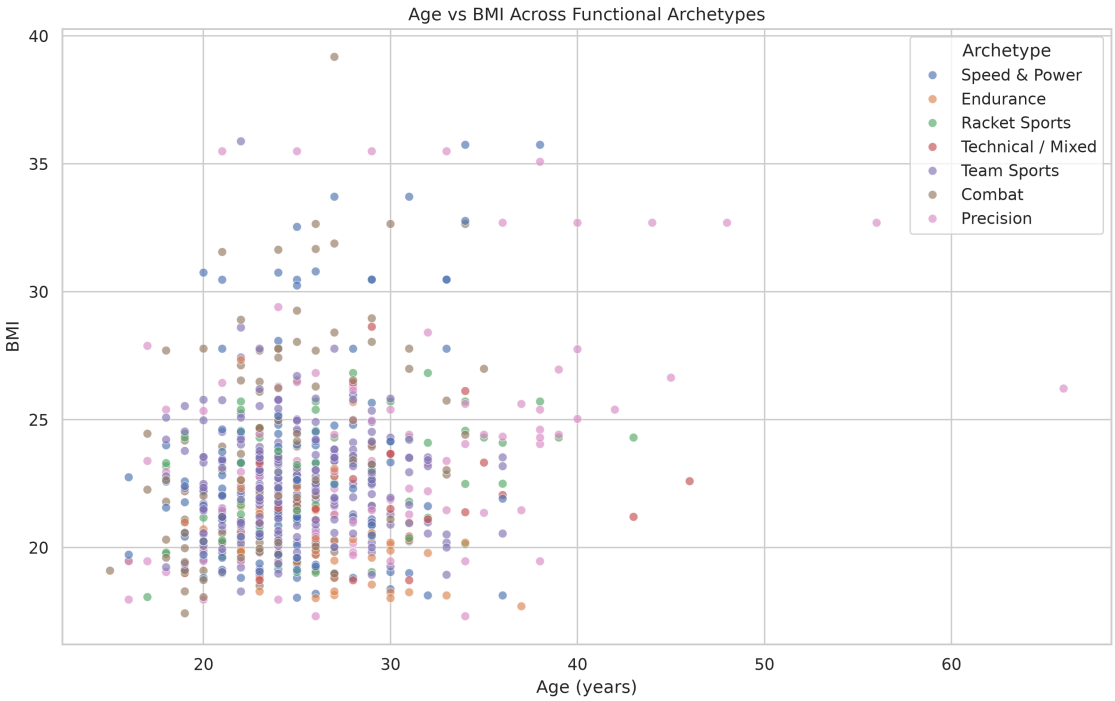


Figure 12: Age–BMI relationship by archetype.

However, the relationship varies between archetypes.

Precision and Combat show moderate positive associations:

$$r_{\text{Precision}} = 0.376$$

$$r_{\text{Combat}} = 0.373$$

These indicate that older observations within these categories tend to have somewhat higher BMI.

Endurance exhibits a negative association:

$$r_{\text{Endurance}} = -0.308$$

Team Sports shows almost no linear relationship:

$$r_{\text{Team Sports}} = -0.043$$

These differences suggest that there is no single consistent age–BMI relationship across all functional archetypes.

The observed correlations are descriptive and cannot determine whether aging itself causes changes in BMI.

14 Limitations of the Analysis

Several factors should be considered when interpreting these results.

1. **Unequal Sample Sizes:**

   Archetype sizes range from 29 observations in Technical / Mixed to 245 in Team Sports, affecting the stability of descriptive estimates.

2. **Unequal Sex Representation:**

   Male observations constitute 579 of 737 observations (78.6%), which may influence pooled anthropometric averages.

3. **Repeated Athletes:**

   Some athletes appear in multiple Olympic editions or archetypes.

   The analysis retains one record per athlete-year-archetype rather than one record per unique athlete.

   Therefore, observations are not necessarily independent.

4. **Broad Archetype Classifications:**

   Each archetype combines sports or events with potentially different physical demands.

   Aggregate results may conceal important sport-specific variation.

5. **BMI Limitations:**

   BMI does not distinguish lean muscle mass from fat mass, making it an imperfect measure of body composition among elite athletes.

6. **Dataset Scope:**

   The conclusions apply to the recorded Indian Olympic athlete observations with available measurements and should not be generalized to all Indian athletes.

15 Overall Interpretation and Conclusion

The descriptive analysis reveals distinct differences in age, height, weight, and BMI across the seven functional archetypes of Indian Olympic athletes.

Precision athletes exhibit the highest mean age (29.21 years), weight (71.57 kg), and BMI (24.07), while Racket Sports athletes have the highest mean height (176.66 cm).

In contrast, Endurance athletes have the lowest mean weight (61.04 kg) and BMI (20.36), indicating a relatively lighter physical profile.

The analysis also reveals substantial differences in variability.

Precision has the greatest age variability, while Speed & Power exhibits the greatest variation in height and weight.

Combat and Precision show comparatively large BMI dispersion, whereas Endurance demonstrates a more concentrated BMI distribution.

Sex-wise comparisons reveal that male athletes have higher average height, weight, and BMI across all archetypes.

However, some key patterns remain consistent within both sexes.

Racket Sports athletes are the tallest on average, and Endurance athletes exhibit the lowest mean weight and BMI in both male and female subgroups.

The height–weight analysis identifies a positive association ($r = 0.698$), while the age–BMI relationship is considerably weaker ($r = 0.240$) and varies across archetypes.

Overall, the findings suggest that the functional categories differ in their observed anthropometric profiles.

These patterns are consistent with the possibility that different sports involve different physical demands.

However, the descriptive analysis cannot establish these demands as the cause of the observed differences.

The findings should also be interpreted in light of unequal sample sizes, sex imbalance, repeated athlete observations, and variation among sports within each archetype.

Final Conclusion:

Functional archetype classification provides a useful descriptive framework for understanding variation in the physical characteristics of recorded Indian Olympic athletes.

The results highlight the importance of examining both central tendencies and within-group variability, as well as accounting for sex composition and sample-size differences when comparing athletic populations.

16 Scope and Interpretive Limitations

This Question 1 report emphasizes descriptive comparisons. Confidence intervals, formal hypothesis tests, and Q–Q plots are not required to answer the stated question and are not presented as inferential evidence. The same athlete may appear in different years or different archetypes, the original height/weight completeness filter may affect representativeness, age imputation can affect summary ages, and sex and category sizes are highly unequal. The included BMI and correlation analyses are exploratory extensions. Multiple linear regression and year-adjusted modelling are addressed by the project's separate Questions 2 and 3, not established by these descriptive correlations.

Data and Reproducibility

Source dataset: *120 Years of Olympic History* (Kaggle), filtered to Indian Summer Olympic entries (1936–2016), using the project's cleaned `data_read_to_use.csv`. The category classifications and main aggregation follow `classification.ipynb` and `stats.ipynb`; figures are sourced from `q1_plots/`. The Athletics-specific figures were computed from the cleaned input with one row retained per athlete, year, and sub-discipline.